

SHUCHANG LIU

RESEARCH EXPERIENCE

Massachusetts Institute of Technology

Cambridge, USA

Postdoctoral researcher

Jun. 2024 – present

- A physically-constrained downscaling approach based on dynamical downscaling and machine learning
[Funded by Swiss National Science Foundation Postdoc Mobility fellowship, CHF 128k]
- Constraining precipitation uncertainty under climate change using machine learning (MIT Climate Grand Challenges Initiative)
- Generalized convection parameterization: adapting current-climate trained models for warmer climates
(Affiliate scientist of Multiscale Machine Learning In Coupled Earth System Modeling)

ETH Zürich

Zürich, Switzerland

Postdoctoral researcher

Oct. 2023 – May. 2024

- How to deal with double-ITCZ when downscaling in the tropics

PhD candidate

Oct. 2020 – Sep. 2023

- The potential of high-resolution climate models in the tropics in constraining climate-change uncertainties

Tsinghua University

Beijing, China

Student Researcher

Sep. 2017 - June 2020

- The impact of climate mitigation pathways on the heatwave and air pollution-related human health in China

EDUCATION

ETH Zürich

Zürich, Switzerland

PhD Atmospheric and Climate Science

Sep. 2023

Tsinghua University

Beijing, China

M.S. Environmental Sciences and Engineering; outstanding graduate

June 2020

Nanjing University

Nanjing, China

B.S. Environmental Engineering; outstanding graduate

July 2017

PRESENTATIONS

Dec. 2025	American Geophysical Union Meeting (Talk), New Orleans, USA
Nov. 2025	Invited talk at Princeton University, USA
Oct. 2025	Invited talk by Machine Learning and Climate Meetups in the Greater Boston area, MIT, USA
Oct. 2025	Invited talk by Multiscale Machine Learning in Coupled Earth System Modeling, NYU, USA
July 2025	Invited talk at University of Lausanne, Switzerland
Dec. 2024	American Geophysical Union Meeting (Poster), Washington, D.C., USA
Dec. 2023	American Geophysical Union Meeting (Talk), San Francisco, USA
April 2023	European Geophysical Union Meeting (Talk), Vienna, Austria
May 2022	European Geophysical Union Meeting (Talk), Vienna, Austria

SERVICES

NSF Review Panelist for GEO directorate

2026

Reviewer for *Earth's Future*, *Journal of Advances in Modeling Earth Systems*, *Geophysical Research Letters*, and *Artificial Intelligence for the Earth Systems*

Since 2025

Organizer of monthly *Machine Learning and Climate Meetups* in the Greater Boston area.

Since 2025

(Engaging participants from MIT, Harvard, Google, Boston University, and Boston College across all career stages.)

TEACHING & SUPERVISING EXPERIENCES

Bachelor's Courses

Risk analysis of environmental health	Tsinghua University
Numerical method in environmental physics, Environmental Systems, Python in Geosciences	ETH Zürich

Master's Courses

Numerical modeling of weather and climate models	ETH Zürich
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Bachelor's Thesis

Visualization of simulated clouds over the Atlantic	ETH Zürich
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Master's Thesis

Extreme Heat Waves Over the Persian Gulf	ETH Zürich
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PUBLICATIONS

H-index: 13, Citations: 809

JOURNAL

Liu, S., Paul A. O'Gorman. CERA: A Framework for Improved Generalization of Machine Learning Models to Changed Climates. *Under revision. Journal of Advances in Modeling Earth Systems*. (2025)
<https://arxiv.org/pdf/2509.00010>

Liu, S., Zeman, C., & Schär, C. Assessing cloud feedbacks over the Atlantic with bias-corrected downscaling. *Journal of Advances in Modeling Earth Systems*, 17, (2025): e2024MS004661. <https://doi.org/10.1029/2024MS004661>

Wang, Y., Liu, Y., **Liu, S.**, Wang, B., Duan, X., Wang, S., & Zhao, B. Changing climate alters the time people spend outdoors. *Sustainable Horizons*, (2025): 13, 100133. <https://doi.org/10.1016/j.horiz.2025.10013>

Liu, S., Zeman, C., & Schär, C. Dynamical Downscaling of Climate Simulations in the Tropics. *Geophysical Research Letters*, 51, (2024): e2023GL105733. <https://doi.org/10.1029/2023GL105733>

Yao, M., Niu, Y., **Liu, S.**, Liu, Y., Kan, H., Wang, S., ... & Zhao, B. Mortality burden of cardiovascular disease attributable to ozone in China: 2019 vs 2050. *Environmental Science & Technology* (2023): 57(30), 10985-10997. <https://doi.org/10.1021/acs.est.3c02076>

Jiang, Y., Ding, D., Dong, Z., **Liu, S.**, Chang, X., Zheng, H., ... & Wang, S. Extreme emission reduction requirements for china to achieve world health organization global air quality guidelines. *Environmental Science & Technology*, (2023): 57(11), 4424-4433. <https://doi.org/10.1021/acs.est.2c09164>

Liu, S., Zeman, C., Sørland, S. L., & Schär, C. Systematic Calibration of a Convection-Resolving Model: Application over Tropical Atlantic. *Journal of Geophysical Research: Atmospheres*, (2022): e2022JD037303. <https://doi.org/10.1029/2022JD037303>

Liu, Y., **Liu, S.**, Wang, S., & Zhao, B. How will window opening change under global warming: A study for China residence. *Building and Environment*, (2022): 209, 108672. <https://doi.org/10.1016/j.buildenv.2021.108672>

Liu, S., Xing, J., Wang, S., Ding, D., Cui, Y., & Hao, J. Health benefits of emission reduction under 1.5°C pathways far outweigh climate-related variations in China. *Environmental Science & Technology*, (2021): 55(16), 10957-10966. <https://doi.org/10.1021/acs.est.1c01583>

Liu, S., Xing, J., Sahu, S. K., Liu, X., **Liu, S.**, Jiang, Y., ... & Wang, S. Wind-blown dust and its impacts on particulate matter pollution in Northern China: current and future scenarios. *Environmental Research Letters*, (2021): 16(11), 114041. <https://doi.org/10.1088/1748-9326/ac31ec>

Ding, D., Xing, J., Wang, S., Dong, Z., Zhang, F., **Liu, S.**, & Hao, J. Optimization of a NO_x and VOC cooperative control strategy based on clean air benefits. *Environmental science & Technology*, (2021): 56(2), 739-749. <https://doi.org/10.1021/acs.est.1c04201>

Sahu, S. K., **Liu, S.**, Liu, S., Ding, D., & Xing, J. Ozone pollution in China: Background and transboundary contributions to ozone concentration & related health effects across the country. *Science of the Total Environment*, (2021): 761, 144131. <https://doi.org/10.1016/j.scitotenv.2020.144131>

Jiang, Y., Xing, J., Wang, S., Chang, X., **Liu, S.**, Shi, A., ... & Sahu, S. K. Understand the local and regional contributions on air pollution from the view of human health impacts. *Frontiers of Environmental Science & Engineering*, (2021): 15(5), 88. <https://doi.org/10.1007/s11783-020-1382-2>

Liu, S., Xing, J., Westervelt, D. M., **Liu, S.**, Ding, D., Fiore, A. M., ... & Wang, S. Role of emission controls in reducing the 2050 climate change penalty for PM_{2.5} in China. *Science of the Total Environment*, (2021): 765, 144338. <https://doi.org/10.1016/j.scitotenv.2020.144338>

Liu, S., Xing, J., Wang, S., Ding, D., Chen, L., Hao, J. Revealing the impacts of transboundary pollution on PM_{2.5}-related deaths in China. *Environment International*, (2020): 134, 105323. <https://doi.org/10.1016/j.envint.2019.105323>

Xing, J., Lu, X., Wang, S., Wang, T., Ding, D., Yu, S., ... & Hao, J. The quest for improved air quality may push China to continue its CO₂ reduction beyond the Paris Commitment. *Proceedings of the National Academy of Sciences*, (2020): 117(47), 29535-29542. <https://doi.org/10.1073/pnas.2013297117>

Chen, L., Xing, J., Mathur, R., **Liu, S.**, Wang, S., & Hao, J. Quantification of the enhancement of PM_{2.5} concentration by the downward transport of ozone from the stratosphere. *Chemosphere*, (2020): 255, 126907. <https://doi.org/10.1016/j.chemosphere.2020.126907>

Sahu, S. K., Chen, L., **Liu, S.**, Ding, D., & Xing, J. The impact of aerosol direct radiative effects on PM_{2.5}-related health risk in Northern Hemisphere during 2013–2017. *Chemosphere*, (2020): 254, 126832. <https://doi.org/10.1016/j.chemosphere.2020.126832>

BOOK

Wang, S., **Liu, S.** Air pollution and lung cancer risks. In: Nriagu, J.O. (Ed.), *Encyclopaedia of environmental health*, 2nd Edition. Elsevier Science. (2019). <https://doi.org/10.1016/B978-0-12-409548-9.11823-8>